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Training package
In

Clinical Immunology

For
Students of fourth class of
Medical Laboratory Techniques Department



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Lecture No. 1

Rheumatoid Arthritis

Rheumatoid Arthritis (RA) is a chronic systemic inflammatory disease of undetermined etiology involving primarily the synovial membranes and articular structures of multiple

joints. The disease is often progressive and results in pain, stiffness, and swelling of joints. In late stages, deformity and ankyloses develop. Women are affected more than men, with a female: male ratio of **3:1**; the disease onset reaches between 35 and 50 years.

Causes

The causes of RA is unknown.

- **Genetic**: Certain **HLA-DR4** molecules associated with RA (e.g. **HLA-DR beta *0401, 0404, or 0405**); in addition, **HLA-DR1** (**HLA-DR beta *0101**) carries this shared epitope and confer risk, particularly in certain southern European areas.
- **Environmental:** for many decades, numerous infection age bnts have been suggested to induce RA. Among these are Mycoplasma organisms, Epstein Barr virus, rubella virus, cytomegalovirus and herpes virus.
- **Hormonal:** Sex hormones may play a role, as evidenced by the disproportionate number of females with RA, its amelioration during pregnancy, its recurrence in the early postpartum period, and its reduced incidence in women using oral contraceptives. Hyperprolactinemia may be a risk factor for RA.

Immunopathogensis

Rheumatoid Arthritis is a disease result from immunological response to an antigen within the joint. This antigen could be self molecules expressed in the joint or could be foreign (e.g. bacterial or viral) antigen sequestered in joint tissue. The nature of this immunological response and the target antigen remain uncertain.

1- Unknown antigen stimulates the activation of T lymphocytes that, in turn activate synovial macrophages.

2- The macrophages secrete the pro-inflammatory cytokines, TNF-α and IL-1, which activate osteoclasts and chondrocytes. This "two-pronged attack" results in the destruction of cartilage and bone.

3- The chondrocytes begin to produce high quantities of fibroblast growth factor and GM-CSF, which completes a harmful cycle that can result in reactivation of the macrophage.
4- Also, B cells are activated by polyclonal stimulation and produce immunoglobulins, especially rheumatoid factors, that stimulate the activation of complement through

immune complex. Moreover, proinflammatory cytokines, especially TNF-α and IL-1, lead to the increased of proliferation and activation of fibroblast. This result in synovitis with pannus formation and in consecutive bone and joint damage.

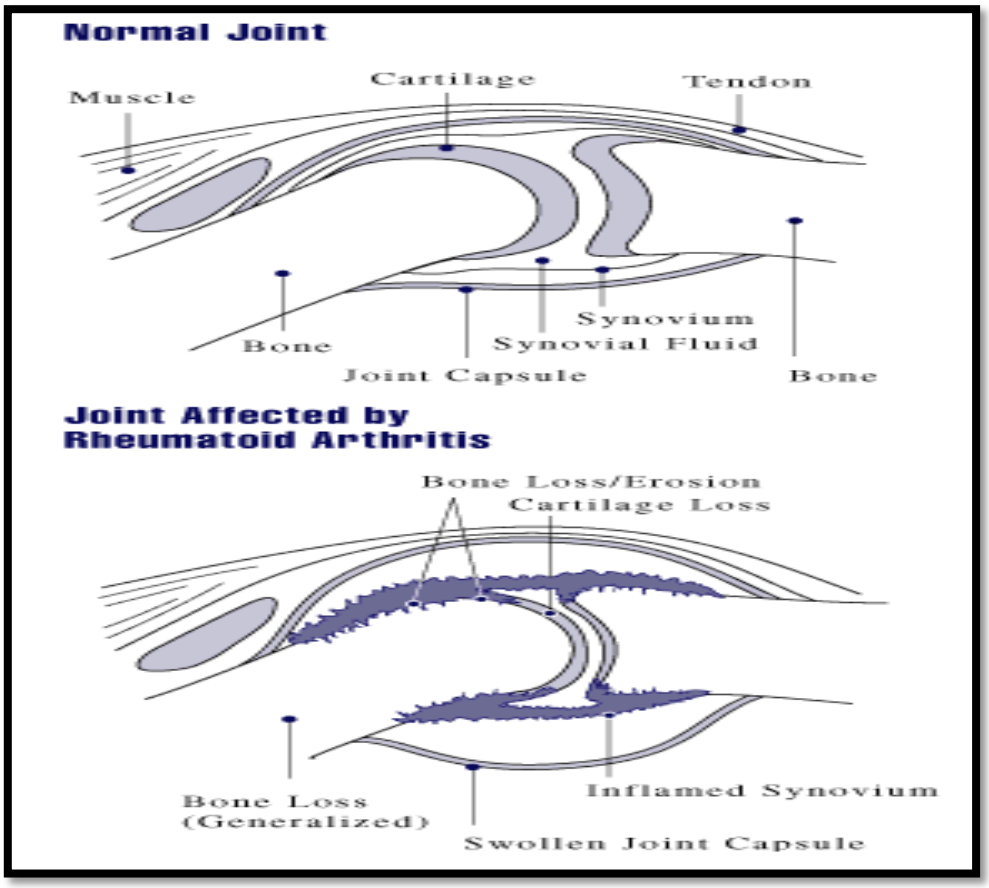
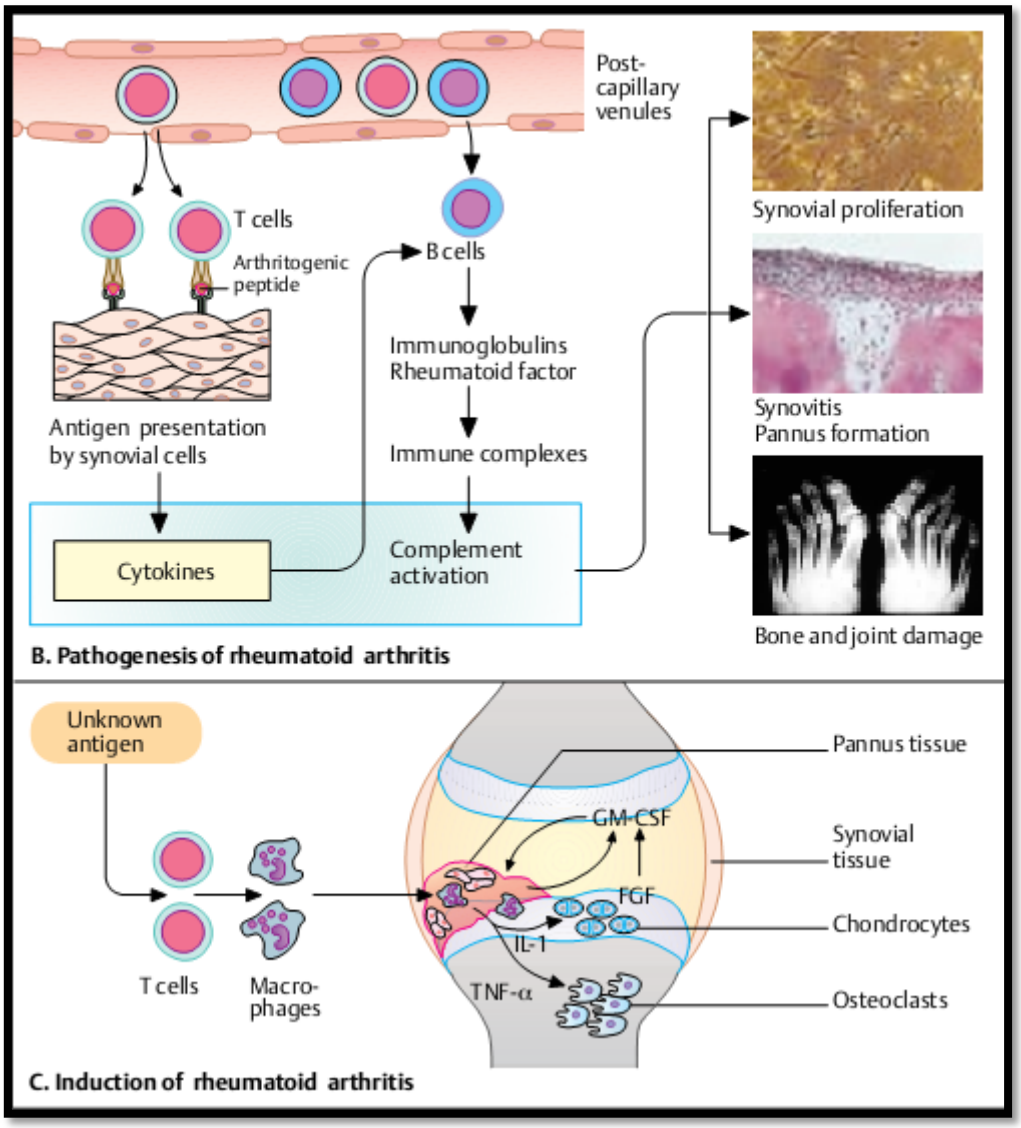


Figure: Comparison between normal and RA joint



Symptoms of RA Patients

RA usually affects joints on both sides of the body equally. Wrists, fingers, knees, feet, and ankles are the most commonly affected. The disease often begins slowly, usually with only minor joint pain, stiffness, and fatigue.

Other symptoms include:

- Chest pain when taking a breath (pleurisy).
- Pericarditis.
- Eye burning, itching, and discharge.
- Nodules under the skin (usually a sign of more severe disease).
- Burning in the hands and feet.
- Sleep difficulties.

Joint symptoms may include:

- Morning stiffness, which lasts more than 1 hour, is common. Joints may feel warm, tender, and stiff when not used for an hour.
- Joint pain is often felt on the same joint on both sides of the body.
- Over time, joints may lose their range of motion and may become deformed.

Laboratory diagnosis

A. **Synovial fluid analysis**

1. Inflammatory synovial fluid (WBC count > 2000/µL) is present with WBC counts generally from 5,000-50,000/µL.
2. Usually, neutrophil predominance (60-70%) is observed in the synovial fluid(in contrast with mononuclear cell predominance in the synovium). Because of a transport defect, the glucose levels of pleural, pericardial, and synovial fluids in patients with RA are often low compared to serum glucose levels.

B. **Immunological diagnosis** include autoantibodies (e.g., **RF, anti-RF33 (nuclear antigen), anti-CCP, ANA)**.

1. **Rheumatoid factor(RF)** refers only to the IgM antibody which binds aggregated IgG as antigen. During the first year of illness, rheumatoid factor is more likely to be negative with some individuals converting to seropositive status over time. RF is also seen in other illnesses, for example Sjögren's syndrome, Hepatitis C, chronic infections and in approximately 10% of the healthy population, therefore the test is not very specific.
2. **Anti-cyclic Citrullinated peptide** (**anti-CCP**) is the highly sensitivity (90-96%) for RA, can identify RA years before symptoms develop and is the most specific test for RA.
3. **Antinuclear antibodies (ANA)** are present in approximately 30% of patient with RA. This test is not routinely performed in the early disease.
4. **C-reactive protein (CRP)** for acute active arthritis.

Table: autoimmune response identified in patients with RA.

Autoantigen	**Antibodies in RA**	**T cell response in RA**	**Specificity for RA**
IgG	Yes: RF		No
Type II collagen and other cartilage antigens	Yes: in10-20%	Yes: in10-20%	No
Citrullinated proteins (CCP)	Yes	Probably	Yes

C. **Hematological tests**

1. **Complete blood count (CBC)** indicate the presence of anemia in normocytic and normochromic.
2. **Thrombocytosis** may be present.
3. **Erythrocyte sedimentation rate (ESR)** is elevated in approximately 90% of patient with RA. This test is not routinely performed in the acute setting

Table 10.5 Some laboratory findings in active rheumatoid arthritis (RA) contrasted with those in active systemic lupus erythematosus (SLE).

	RA	SLE
Rheumatoid factor	Positive in 70%	Positive in 30%
• Titre	High	Low
Antinuclear antibody		
• Class	IgM	IgG
• Titre	Low	High
• Proportion of patients positive	40%	95%
DNA binding		
• Proportion of patients positive	<10%	70–85%
C3 + C4 levels	Normal or ↑	↓ or normal
C-reactive protein	↑	Usually normal
Immunofluorescent examination of a skin biopsy*	–ve	+ve
Serum IgG levels	Usually normal	Often ↑

Lecture No. 2

Systemic lupus erythematosus

Systemic lupus erythematosus (SLE) is a systemic autoimmune disease characterized by the production of autoantibodies and a diversity of clinical manifestations. It most commonly presents in women during their child-bearing years, in which the immune system targets intracellular particles that contain both nucleic acids and nucleic acid binding proteins.

Etiology and Pathogenesis

Although the etiology of SLE is unknown, multiple factors are associated with the development of SLE, including genetic (HLA- DR2/DR3), racial, hormonal, immune abnormalities and environmental factors (ultraviolet light, viral infection involving molecular mimicry between organism and self for example anti-Sm autoantibody react with p24 gag protein of retroviruses and that anti- Ro recognizes a nucleocapsid protein on vesicular stomatitis virus). One proposed mechanism for the development of autoantibodies involves a defect in apoptosis or clearance of apoptotic cells, leading to a disturbance in immune tolerance. The redistribution of cellular antigens during apoptosis leads to a display of cytoplasmic and nuclear antigens on the cell surface, enhancing immune reactivity to antigens, which are normally protected intracellularly. Activation of antigen-presenting cells by IFN-α might promote presentation of autoantigens to self-reactive T cells. Immune complexes form in the microvasculature, leading to complement activation and inflammation. Antibody–antigen complexes deposit on the basement membranes of skin and kidneys. In active SLE, this process has been confirmed based on the presence of complexes of nuclear antigens such as DNA, immunoglobulins, and complement proteins at these sites.

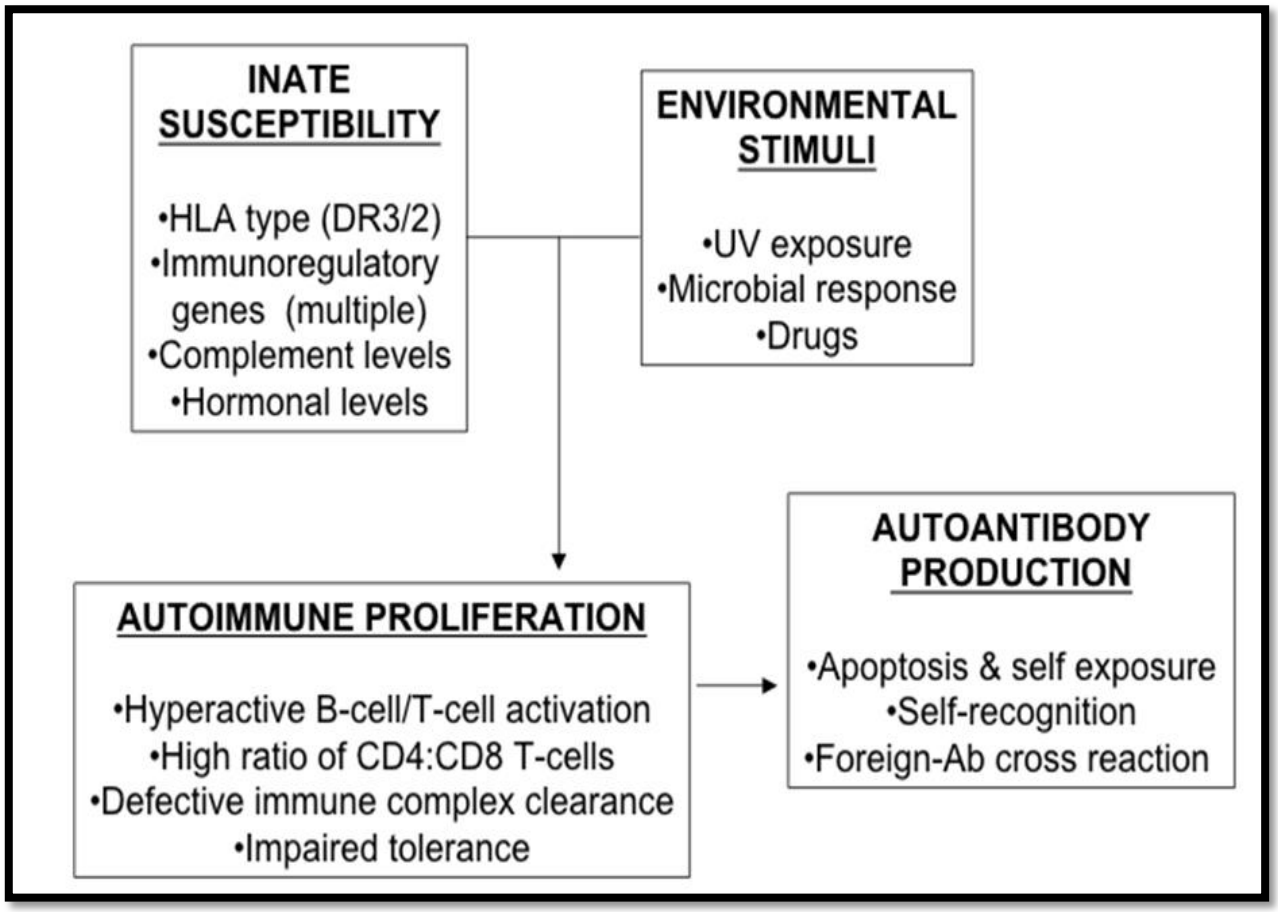
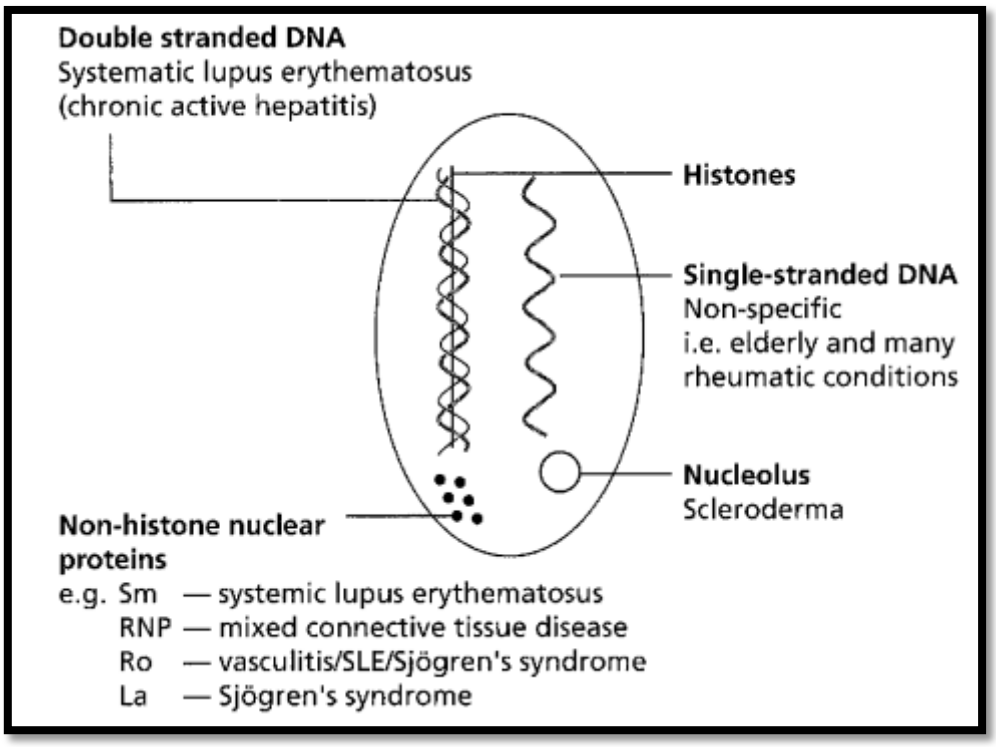


Figure: Etiology of SLE.



Nuclear antigens

Clinical Features

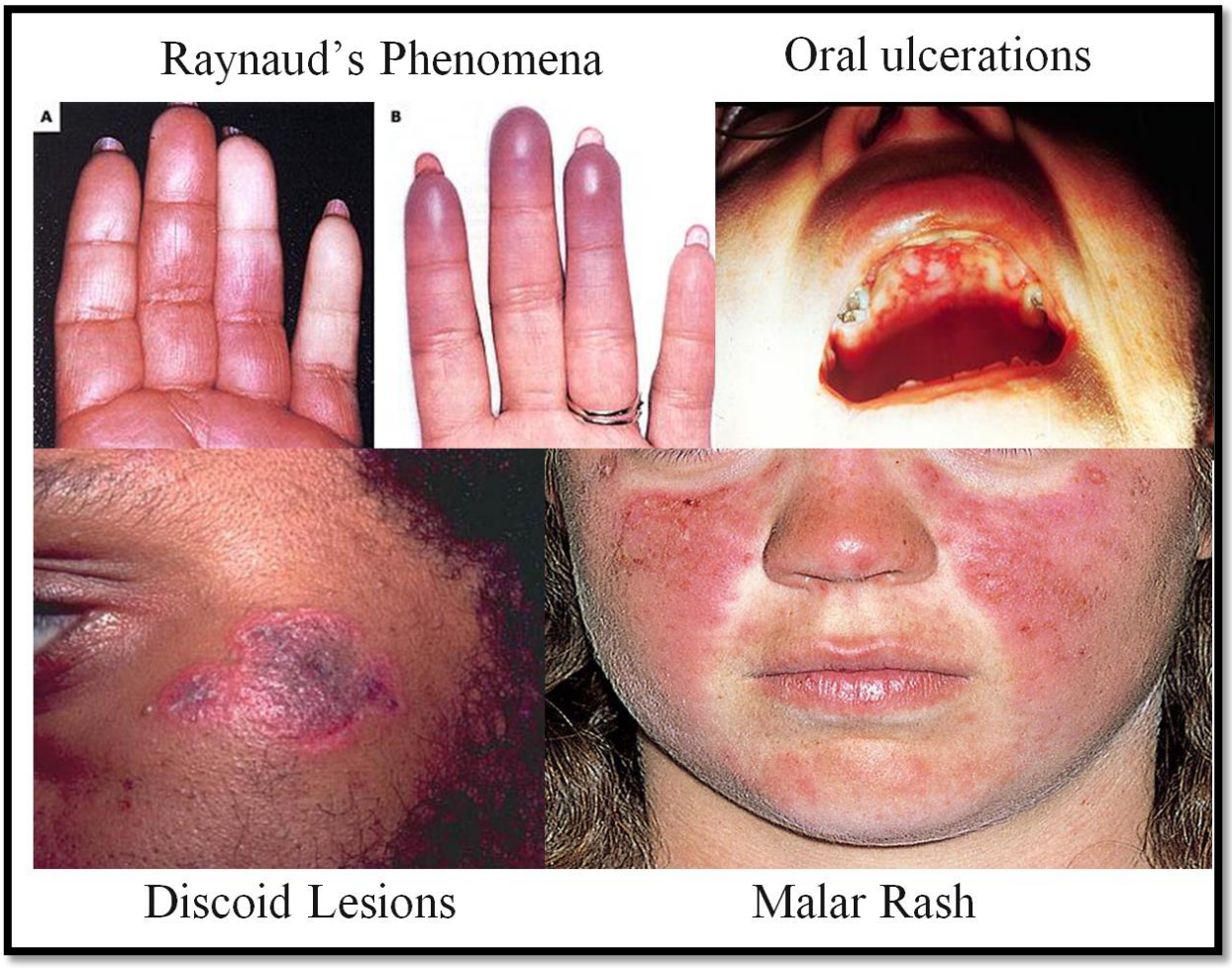
Systemic lupus erythematosus is a multisystem disease and can affect virtually all organs and system; whilst some manifestations are common, others are rare. Therefore, joint, skin and blood are affected in 80-100% of patients, kidneys, CNS and cardiopulmonary system in over 50%; while thrombosis, a typical lupus manifestation associated with possession of the anticardiolipin antibody, is present in 10% of patients. Systemic manifestations including fatigue, malaise, fever, anorexia, nausea and weight loss, are present in the great majority of patients.

The symptoms difference according to the infected organ and including arthritis, arthralgia, malar rash, an erythematous rash over the nasal bridge, photosensitivity, discoid lesions, headache, migraine, nephrotic syndrome, pleuritis and pericarditis.

Table 10.12 American College of Rheumatology criteria for diagnosis of systemic lupus erythematosus (SLE).*

Malar rash
Discoid rash
Photosensitivity
Oral ulcers
Non-erosive arthritis
Serositis (pleuritis/pericarditis)
Renal disease (persistent proteinuria/urinary casts)
Neurological disorder (seizures/psychosis)
Haemolytic anaemia/leucopenia/lymphopenia/thrombocytopenia
Antinuclear antibody
Antibodies to dsDNA/antibodies to extractable nuclear antigens/ antiphospholipid antibodies

* To establish a diagnosis of SLE a patient must have four or more of these criteria.



General laboratory findings

The most frequent laboratory alteration that is identified is normochromic normocytic anemia of chronic disorders. Occasinally a Coombs-positive haemolytic anemia is observed. Leukopenia (probably autoantibody mediated), especially lymphopenia, and thrombocytopenia are frequent. The erythrocyte sedimentation rate is typically elevated, while C-reactive protein tends to be normal. Urinalysis can show haematuria, proteinuria and renal casts in the presence of glomerulonephritis.

Immunological laboratory findings

All patients in whom SLE is suspected should be tested for antinuclear antibodies, including those to dsDNA and extractable nuclear antigens (ENA), and for antiphospholipid antibodies, as well as having their serum level of IgG and complement components, C3 and C4 measured, Antihistone antibodies are also present in patients with drug-induced SLE, most frequently associated with hydralazine and procainamide therapies.

The sero-immunological hallmark of SLE is **antinuclear antibodies (ANA)**, in the absence of ANA, the diagnosis of SLE is put into question, even though some 5% of patients may have an ANA-negative serology.ANA is currently detected by an indirect immunofluorescence technique, where diluted patients serum is applied to frozen tissue, especially liver, of rodent origin and cell lines of human origin, such as the HEp2 cell line derived from a laryngeal tumor, in which nuclei are prominent, are used as substrate to detect ANA. If the patient is ANA positive, the autoantibody will bind to nuclei. To reveal this binding, a second antibody tagged with fluorescent label is add. This second antibody will bind and ANA will then be seen by placing the preparation under a fluorescence microscope. Four patterns of fluorescence can be seen indicating different types of antinuclear antibodies.

Table. Immunofluorescence Patterns of Antinuclear Antibodies

Pattern	Antigen	Disease association(s)
Peripheral	Double-stranded DNA	SLE
Homogeneous	DNA-histone complexes	SLE and other connective tissue diseases
Speckled	Non-DNA nuclear antigens	
	Sm	SLE
	ribonucleoprotein	Mixed connective tissue disease, SLE, scleroderma, etc.
	SS-A, SS-B	Sjögren's disease
Nucleolar	Nucleolus-specific RNA	Scleroderma

ANA is a very sensitive test for SLE, being present in virtually all patients and frequently at high titers; its disease specificity is relatively low since it is frequently found in other rheumatic diseases, as well as in autoimmune liver disease, during viral infections and, occasionally, at low titers, in normal subjects.

DNA antibodies are the most important in SLE. They can react with **single-stranded DNA (ssDNA) or with double-stranded DNA (dsDNA)**.Although anti-ssDNA may be found in many diseases besides SLE, anti-dsDNA autoantibodies are found almost

exclusively in SLE (70%of the patients). While the disease specificity of dsDNA autoantibodies is high, that of ANA is low.

Anti-dsDNA autoantibodies are usually detected by very analytically sensitive technique, such as radioimmunoassay (RIA) or enzyme linked immunosorbent assay (ELISA).they can also be detected by immunofluorescence staining of an organelle called a kinotoplast in the flagellate *Crithdia luciliae*,which contains dsDNA.

In a patient with lupus nephritis, a kidney biopsy is frequently obtained for diagnostic reasons the glomeruli of such bioptic renal material contain antigen-antibody complexes. By applying a fluorescent antibody directed against human antibody (similar to that used in the second step of ANA detection) to frozen section of the kidney biopsy. This one step technique is known as direct immunofluorescence.

Extractable nuclear antigens (ENA) include **Sm** (Smith), **RNP** (ribonucleoprotein), **Ro** (Robert) also called SS-A (Sjogrens syndrome antigen A) and anti- **La** (Lane) or SS-B (Sjogrens syndrome antigen B).

Anti-ENA antibodies are include anti-Sm found almost exclusively in SLE, and anti-RNP more typically associated with mixed connective tissue disease than with SLE, anti- Ro and anti-La are found in Sjogrens syndrome. Other Anti-ENA antibodies are **anti-Jo-1**, anti **Scl-70** and **anticentromere**, which are associated mainly with polmyositis, systemic sclerosis and CREST syndrome respectively. Anti-ENA antibodies are normally detected by immunodiffusion or ELISA technique.

The lupus anticoagulant causes a prolonged clotting time in vitro but thrombosis in vivo. It is often found in associated with other autoantibodies to phospholipids, such as anticardiolipin antibodies and false positive tests for syphilis.

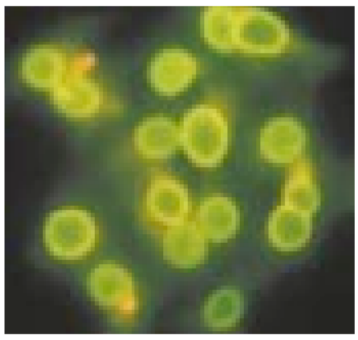
Assessment of the complement profile is of importance in management. Serial determinations of CH_{50}, a functional assay measuring complement hemolytic activity, and of the individual factors C3 and C4, inform on how much immune complexes are consuming complement.

Table 10.13 Laboratory findings in untreated systemic lupus erythematosus (SLE)*.

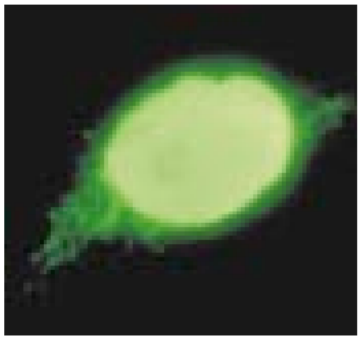
Immunological test	%	Haematological	%	Others	%
dsDNA binding	70–85	Raised ESR	60	C-reactive protein–normal unless infection present	
Antinuclear bodies (high titre; IgG class)	95	Leucopenia	45	Proteinuria	30
Raised serum IgG level	65	Direct Coombs' test positive	40		
Low serum complement C3/C4 levels	60	Lupus anticoagulant	10–20		
Platelet antibodies	60				
Cryoglobulinaemia	60				
Antibodies to ENA:					
Sm	30				
RNP	35				
Ro	30				
La	15				
Antibodies to phospholipids	30–40				
Rheumatoid factor (low titre)	30				
Skin biopsy IgG, C3 and C4 deposits in normal skin	75				

* Figures show percentage of patients with positive tests.

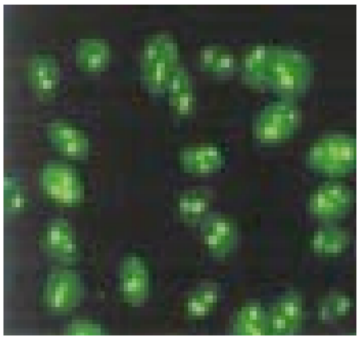
ESR, erythrocyte sedimentation rate; ENA, extractable nuclear antigens; RNP, ribonucleoprotein.



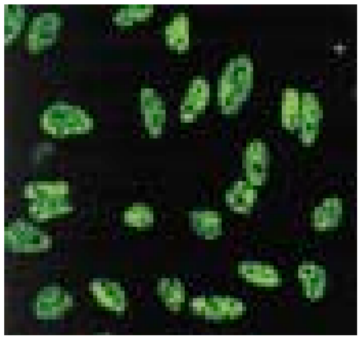
1. Rim pattern (150x magnif; anti-DNA)



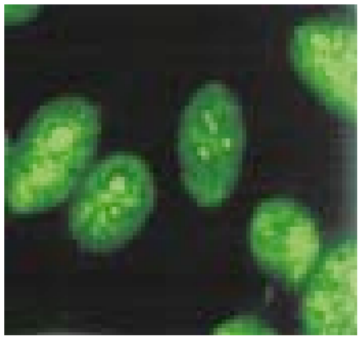
2. Homogenous pattern (435x enlarged; anti-DNA)



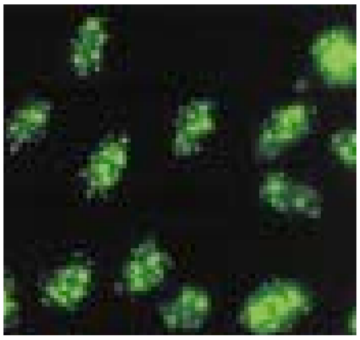
3. Nucleolar pattern (e.g. fibrillarin)



4. Coarse-speckled pattern (U1RNP/Sm)



5. Fine-speckled pattern (Ro/La)



6. Anti-centromere antibody pattern

Lecture No. 3

Ankylosing spondylitis

The term spondyloarthritis (SpA) (otherwise known as spondyloarthropathy) encompasses a heterogeneous group of inflammatory diseases characterized by spinal and peripheral joint oligoarthritis, inflammation of the attachments of ligaments and tendons to bones (enthesitis) and, at times, mucocutaneous, ocular, and/or cardiac manifestations. These disorders show familial aggregation and are typically associated with genes of the major histocompatibility complex (MHC), particularly human leukocyte antigen (HLA)-B27 The SpA include: (1) ankylosing spondylitis (AS); (2) reactive arthritis (ReA)—known previously as Reiter's syndrome; (3) psoriatic arthritis (PsA) and/or spondylitis; (4) enteropathic arthritis and/or spondylitis associated with the inflammatory bowel diseases (IBD), ulcerative colitis (UC), or Crohn's disease; and (5) undifferentiated SpA, which encompasses patients expressing elements of, but failing to fulfill, accepted criteria for one of the above diseases. In addition, isolated acute anterior uveitis (AAU)1 and spondylitic heart disease (complete heart block and/or lone aortic regurgitation) associated with HLA-B27 may also be classified within the spectrum of SpA.

Ankylosing spondylitis (AS) is a chronic inflammatory condition of the spine and sacroiliac joints. It is progressive disease in which restriction of movement is associated with intervertebral ossification of the ligaments. Men, usually below the age of 40, man develop the disease three times more frequently than women. Approximately 90% of the patients are HLA-B*27 positive, while the prevalence of this antigen in the general population is 7%. Of all the adult HLA-B*27- positive individuals, 1-2% have ankylosing spondylitis.

Etiology of AS

The etiology of the disease is unknown, but persistence of specific antigens of the infecting organisms has been demonstrated in these patients. This has led to suggestion that AS is also triggered by infection (possibly in the gastrointestinal tract) in susceptible HLA-B*27- positive individuals. Inflammation occurs and persists in different organs and joints in Ankylosing Spondylitis. Each individual tends to have their own unique pattern of presentation and activity of the illness. The initial inflammation may be a result of an activation of body's immune system, perhaps by a preceding bacterial infection or a combination of infectious microbes. Once activated, the body's immune system becomes unable to turn itself off, even though the initial bacterial infection may have long subsided. Chronic tissue inflammation resulting from the continued activation of the body's own immune system in the absence of active infection is the hallmark of an inflammatory autoimmune disease.

Clinical features

The onset of AS tends to be insidious with a dull lumbar pain; this persists over 3 months and is accompanied by morning stiffness relieved by exercise. Arthritis of the peripheral joints is seen in one third of the patients. Amongst extrarticular manifestation, iritis is the most troublesome: it tends to be unilateral and accompanied by photophobia pain. Inflammation of the colon and ileum is frequent but usually asymptomatic.

Criteria of AS Classification

- **Inflammatory Spinal Pain**: History or present symptoms of spinal pain in back, dorsal or cervical region with at least 4 of the following: **A.** Onset at age < 45 years. B. Insidious onset. **C.** Improved by exercise. **D.** Associated with morning stiffness **E.** at least for 3 months duration
- **Synovitis**: Asymmetric or Predominantly in the lower limbs**.** and one of the followings:
  - Positive family history
  - Psoriasis
  - Inflammatory bowel disease.
  - Alternating buttock pain.
  - Enthesopathy
  - Acute diarrhea
  - Urithritis
  - Sacroiliiatis

The diagnosis of Ankylosing Spondylitis is based on:

- Evaluating the patient's symptoms include pain and morning stiffness of the spine and sacral areas with or without accompanying inflammation in other joints, tendons, and organs.
- A physical examination: the Schober's test is a useful clinical measure of flexion of the lumbar spine performed during examination. Flexibility of the low back and/or neck can be decreased.
- X-ray findings
- Blood tests: Patients with AS tend to have elevated levels of IgA and, when the disease is active, elevated erythrocyte sedimentation rates and levels of C-reactive protein. Rheumatoid factor and antinuclear antibody are consistently negative. The clinical need to assess the HLA-B*27 status of a patient with symptoms and signs of AS is controversial.

Psoriatic Arthritis

Psoriatic Arthritis (PsA) is a chronic and inflammatory arthritis in association with skin psoriasis, characterized by osteolysis and bony proliferation. PsA is classified as one of the subtypes of spondyloarthropathies. Males and females are equally affected. PsA can range from mild nondestructive disease to a severely rapid and destructive arthropathy.

Clinical manifestations include skin and nail psoriasis, dactylitis, enthesitis, osteoperiostitis, large joint oligoarthritis, arthritis mutilans, sacroiliitis, spondylitis and distal interphalangeal arthritis.



Comorbidities in PsA Patients

- Ocular inflammation (Iritis/Uveitis/ Episcleritis).
- Irritable bowel disease (IBD).
- Metabolic Syndrome (Hyperlipidemia, Hypertension, Insulin resistent, Diabetes,Obesity) lead to Higher risk of Cardiovascular disease (CVD)
- Psychosocial burden, Reactive depression, Higher suicidal ideation and Alcoholism.

Two percent of patients with psoriasis develop psoriatic arthropathy; this may affect the peripheral joints or the spine. The psoriasis generally precedes the arthritis by many years; in rare cases where the arthritis comes first, diagnosis may be difficult. A family history of psoriasis is a helpful diagnostic clue and the characteristic nail changes of psoriasis are present in 80% of patients with psoriatic arthritis. Dactylitis – inflammation of an entire digit to look like a sausage – is a distinctive feature. Usually rheumatoid factor (RF) negative and ACPA negative. Radiographic damage can be noted in up to 47% of patients

at a median interval of two years despite clinical improvement with standard DMARD therapy. Treatment is similar to that for RA, including the use of anti- TNF drugs. The prognosis is usually good, although severe joint destruction can occur.

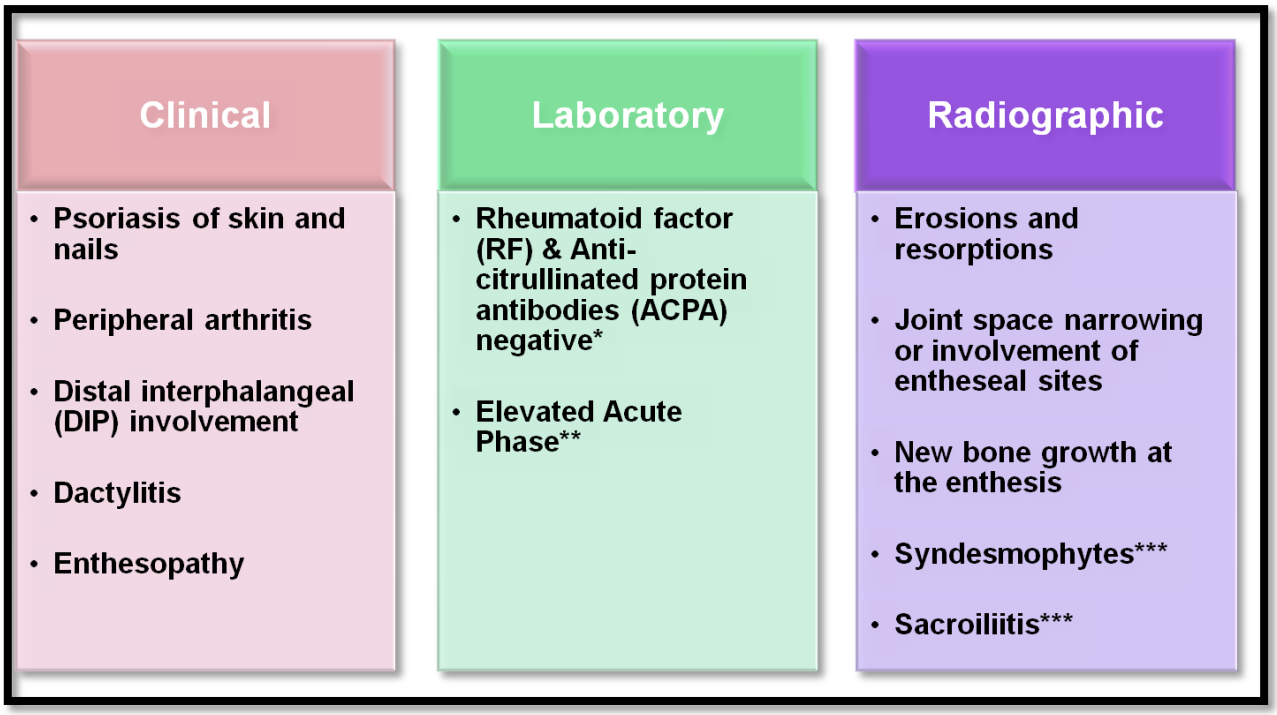


Figure: Main Features of Psoriatic Arthritis

Lecture No. 4

Sjögren's syndrome

Sjögren's syndrome (SS) is a systemic chronic inflammatory disorder characterized by lymphatic infiltrates in exocrine organs (lacrimal and salivary glands). Sjögren's syndrome predominantly effects females (female:male ratio 9:1) in their fourth and fifth decades of life. It was named after the Swedish ophthalmologist Henrik Sjögren's after he reported 19 cases of keratoconjunctivitis in 1933.1 The hallmark feature of SS is deficient tear and saliva production due to lymphocytic infiltration of the salivary and lachrymal glands leading to xerostomia (dry mouth) and xerophthalmia (dry eyes). In addition, SS can involve any organ system and present with a wide spectrum of clinical features.

There are two type of Sjögren's syndrome

- **Primary Sjögren's syndrome** (when the disease occurs alone) is a systemic autoimmune disease that targets the exocrine glands. It is characterized by xerostomia (dry mouth), xerophthalmia (dry eyes).

- **Secondary Sjögren's syndrome** is defined as the former definition of primary Sjögren's in the presence of another autoimmune diseases such as RA, SLE, Scleroderma, and polyarteritis nodosa.

Etiology and Pathogenesis

A number of factors play a role in the development of Sjögren's syndrome

1. Endogenous factors [**HLA-B*8-DRB1*03 ((B8 DR3) and hormones (estrogens)**].
2. Exogenous factors [**viruses (herpes and retrovirus)**].

The pathogenesis of SS is still largely unknown. In a genetically predisposed individual, various environmental factors, such as viral infections, may lead to epithelial cell activation and a protracted inflammatory response with features of autoimmunity. The actual trigger of glandular dysfunction is assumed to be a viral infection. Epithelial cells in the infected gland present viral antigens. This attracts T cells, which infiltrate the glandular tissue and cause a local inflammatory reaction, resulting in damage to glandular tissue. The T cells activate the glandular epithelium and, most importantly, B cells. This results in excessive, uncontrolled B-cell proliferation, which initially manifests in the peripheral blood as hypergammaglobulinemia in association with presence of immune complexes.

Clinical features

The principle feature of Sjögren's syndrome is autoimmune destruction of exocrine gland, most prominently the lacrimal and salivary glands, but also glands at other sites including respiratory mucosa and vagina. The disease commonly affects eyes, mouth, parotid gland, lung, kidney, skin and nerves system. Most individuals with Sjögren's syndrome present with sicca symptoms, such as xerophthalmia (dry eyes), xerostomia (dry mouth)**,** and parotid gland enlargement. Other xeroses such as dry skin and dry vaginal mucosa leading to irritation and dyspareunia may also occur. In addition, numerous extra glandular features may develop, such as arthralgia, arthritis, pulmonary disease, leucopenia, anemia, lymphoadenopathy, neuropathy, renal tubular acidosis and lymphoma.

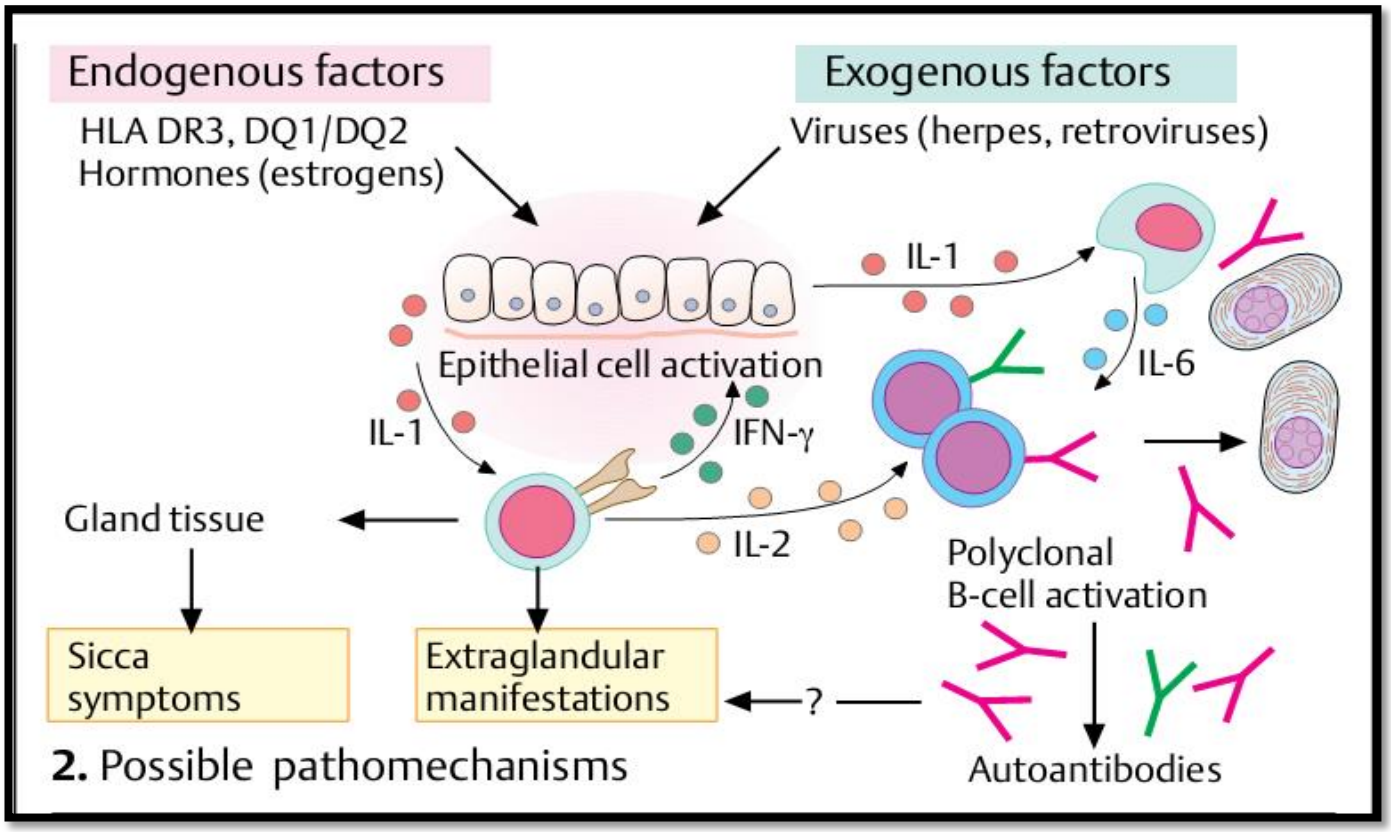
Laboratory finding

The most important diagnostic tests for Sjögren's syndrome are

1. **Schirmer's test:** Lachrymal function can be assessed by the Schirmer's test to quantify the amount of tear production, a slip of sterile filter paper is placed over the lower eyelid, failure to produce sufficient tears within 5 min wet 10 mm of the paper suggest defective tear production, then this is positive for Sjögren's syndrome.
2. **Erythrocyte sedimentation rate (ESR)** is elevated in 80% of patients.
3. **C reactive protein (CRP)** is also elevated in patients.
4. **Rheumatoid factor (RF)** is present in 52% of cases of primary-type Sjögren's syndrome and in 98% of secondary-type cases.
5. **Complete blood count (CBC)** shows anemia and leucopenia.
6. **Autoantibodies**
    - **Anti-SS-A (anti-Ro) and anti-SS-B (anti-La)** are present in most cases of primary-type Sjögren's syndrome. While anti salivary duct antibodies are present in most cases of the secondary type. Anti-Ro/SS-A antibodies are found in over 70% of patients with SS, but are also frequently found in SLE and other autoimmune diseases even in the absence of oral or ocular dryness. Anti-La/SS-B is more specific; it is present in 50% of patients with primary SS or SS/SLE but is rarely seen in other diseases. The pathogenic role of these antibodies is not yet defined except in newborns born to women with anti-Ro/SS-A and/or anti-La/SS-B antibodies. These antibodies can cross the placenta and bind to Ro and La antigens located on the cell surface of fetal myocardial tissue, leading to fetal heart block.
    - **Antinuclear antibodies (ANA)** of the speckled type are present in most cases of primary Sjögren's syndrome.
    - In recent years, research has focused on identifying antibodies more specific for SS, such as anti-a-fodrin and anti-muscarinic acetylcholine receptor antibodies, but the results have been controversial. The major stimulus for saliva production is the binding of acetylcholine to muscarinic acetylcholine receptors. The hypothesis that oral and ocular dryness could result from

antibodies antagonizing the muscarinic acetylcholine receptor-3 is intriguing.

7. **Creatinine clearance** may be diminished in up to 50% of patients.
8. **Rose Bengal, fluorescein and lissamine green staining** are performed on an outpatient basis to the detection corneal and conjunctival damage due to dryness.
9. **Salivary gland evaluation** is done by collection of unstimulated saliva or salivary scintigraphy.



Lecture No. 5

Behcet's Disease

Behcet's disease is a rare, chronic, lifelong disorder that involves **inflammation of blood vessels** throughout the body. It is a form of vasculitis that can lead to ulceration and other lesions, slightly affecting more men than woman characterized by a triad of symptoms, including aphthous ulceration of the oral mucous membranes and genitalia and uveitis. **Rare manifestations** include oligoarthritis of the lower extremities, vasculitis of the pulmonary vessels, cerebrovascular symptoms. **The etiology** of the Behcet's disease is unkown. It is believed to be partly genetic, an association with HLA-B51 has been found.

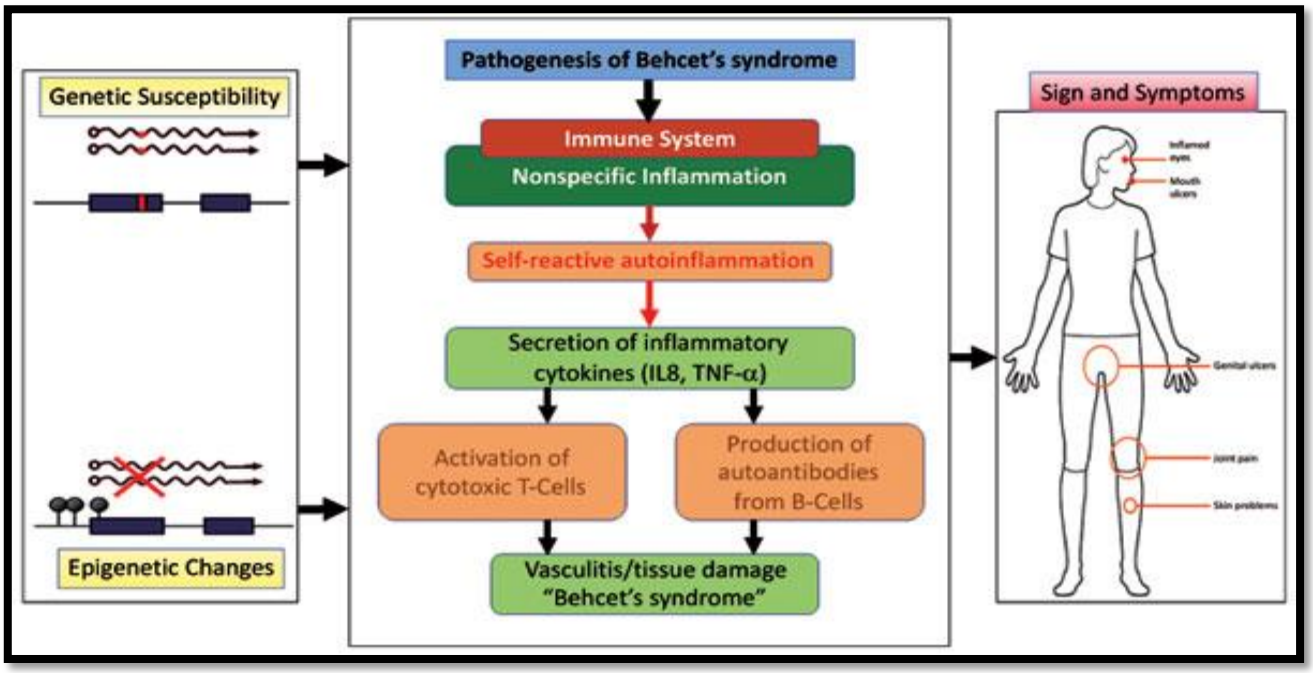


Figure: Pathogencity of Behcet's disease

Risk factors

- **Age.** Behcet's disease most commonly affects men and women in their 20s, 30s and 40s, though children and older adults also can develop the condition. When the condition occurs at an earlier age, it tends to be more severe.
- **Geography.** Although the disease occurs worldwide, people from countries in the Middle East and Asia, including Turkey, Iran, Iraq, Japan and China, are more likely to develop Behcet's.
- **Sex.** While Behcet's disease occurs in both men and women, the disease is usually more severe in men.
- **Genes.** Having certain genes HLA-B51 is associated with a higher risk of developing Behcet's.

Clinical features

- **Mouth:** Painful mouth sores, identical to canker sores, are the most common sign of Behcet's disease. Sores begin as raised, round lesions in the mouth that quickly turn into painful ulcers. The sores heal usually in seven to 21 days, though they do recur.
- **Skin:** Skin lesions may occur in people with Behcet's disease. Skin problems can vary. Some people may develop acne-like sores on their bodies. Others may develop nodules on the lower legs.
- **Genitals:** People with Behcet's disease may develop sores on their genitals. The sores most commonly occur on the scrotum or the vulva.
- **Eyes:** Behcet's disease may cause inflammation in the eye — a condition called uveitis In people with Behcet's disease, uveitis causes redness, pain and blurred vision in one or both eyes

- ❖ **Joints:** Joint swelling and pain most commonly affect the knee in people with Behcet's disease.
- ❖ **Vascular system:** Inflammation in veins and large arteries may occur in Behcet's disease, causing redness, pain and swelling in the arms or legs when a blood clot results.
- ❖ **Digestive system:** Behcet's disease may cause abdominal pain, diarrhea or bleeding.
- ❖ Brain: Inflammation in the brain and nervous system that leads to headache, fever, disorientation, poor balance or stroke.

Diagnosis

There is no diagnostic test and the diagnosis is entirely clinical. Laboratory findings are nonspecific and reflect the inflammatory state, C-reactive protein levels, erythrocyte sedimentation rate (ESR), leukocyte count, complement components, and acute-phase reactants may all be elevated during an acute attack. Internationally accepted diagnosis criteria have been published recently.

International Clinical Criteria for Behcet's Disease

An international group of physicians has established a set of guidelines to aid in the classification of Behcet's patients. The International Clinical Criteria for Behcet's Disease classification states patients must present with:

- Recurrent oral ulcerations (apthous or herpetiform) at least three times in one year.
- Additionally, patients must present any two of the following:
  - Recurrent genital ulcerations.
  - Eye lesions (uveitis or retinal vasculitis) observed by an ophthalmologist.
  - Skin lesions (a variety of rashes or acne-like sores) may be caused by Behcet's disease. Positive
  - Positive pathergy test, your doctor inserts a sterile needle into your skin and then examines the area two days later. If the pathergy test is positive, a small red bump forms under your skin where the needle was inserted. This indicates your immune system is overreacting to the minor injury.

Lecture No. 6

Celiac disease

Celiac disease (CD), an immune-mediated mucosal disorder primarily affecting the small intestine in genetically susceptible individuals, is triggered by the ingestion of dietary gluten. Gluten is the alcohol-soluble protein component of the cereals wheat, rye and barley. It is composed of 2 major protein fractions: glutenin and gliadin; most of the toxic activity exerted by gluten in CD is due to gliadin.